TEXORA · LECTURE SHEET

Sewing Problems and Remedies

A diagnostic method for the sewing room: skipped stitches, thread breakage, unbalanced tension, the three kinds of seam puckering, ply shift and needle damage — each with its cause chain and its remedy.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 4 · LEVEL 3 TERM 1

1. Method: read the symptom, then the system

Sewing faults are diagnosed, not guessed at. Every fault has a short list of candidate causes in five groups — **needle, thread, machine settings, material, operator handling** — and the trained technologist works through them in the order of cheapness: change the needle first (it is the cheapest part and the commonest culprit), re-thread completely (a partly wrong path imitates tension faults), then adjust settings, then suspect the material. This sheet organises the classic faults in that diagnostic order.

2. Stitch-formation faults

FAULT	CAUSE CHAIN	REMEDY
Skipped stitches	Loop not caught: bent/blunt needle, needle too fine for thread, wrong needle–hook timing (loop lift), flagging of light fabric under the foot, thread too elastic.	New needle of correct system/size; check hook timing and loop lift (1.8–2.2 mm); hold-down pressure or smaller needle-hole; correct thread.
Thread breakage (needle thread)	Burrs on needle eye/hook/plate, excessive tension, thread too weak or too thick for the eye, needle overheating on synthetics, sharp thread guides.	Polish/replace burred parts; reduce tension to the minimum that balances; match thread–needle sizes; silicone lubricant/needle cooling; check entire thread path by hand.
Bobbin thread breakage	Bobbin overfilled or unevenly wound, damaged bobbin case, excessive bobbin tension, lint packed under the spring.	Rewind correctly (even, below the rim); clean and reset the case spring; replace a scored case.
Unbalanced stitch	Knot visible on top (needle tension low / bobbin high) or below (reverse).	Set bobbin tension first as the reference, then adjust needle tension until the knot buries mid-ply on the actual fabric.

Stitch density varies	Feed dog worn or set low, presser pressure low, operator pulling/pushing the work.	Reset feed dog height (~0.8–1.0 mm above plate); increase pressure; retrain handling — the machine feeds, the operator guides.

3. Seam-appearance faults

FAULT	CAUSE CHAIN	REMEDY
Seam puckering	Three kinds: tension pucker (threads too tight, seam contracts), feed pucker (plies fed unequally, lower stretched), displacement/shrinkage pucker (fabric structure jammed by dense stitching, or thread shrinks in pressing/washing).	Diagnose by cutting threads of a test seam: if the fabric relaxes flat, tension pucker; if one ply stays longer, feed pucker. Reduce tensions, lower stitch density, match presser/feed settings, use finer thread and needle, sew with the grain direction that feeds evenly.
Ply shift / mismatched ends	Drop feed drives the lower ply, friction drags the upper; long seams accumulate the difference.	Needle feed or walking foot; notches matched by the operator; reduced presser pressure; hold both plies with even hand tension.
Wavy / fluted knit edges	Edge stretched by feeding or by looper tension.	Differential feed ratio below 1; reduce looper tensions; support the edge; check presser pressure.
Needle damage (holes, cuts)	Wrong point (sharp in knits), blunt needle, needle too coarse, excessive density.	Ball point for knits; size down; renew needles on schedule (per shift/per style), not on failure.
Oil stains	Over-oiled hook, leaking seals, threading with oily hands.	Correct oil level and wicking; run off on waste after oiling; clean product handling.

4. A worked diagnosis

Case. A shirt line reports puckered side seams only on the new poplin style, although the same machines sewed the previous twill acceptably. Working the groups: needle — unchanged size 80 sharp, plausible on both fabrics; thread — same tex 24; settings — tensions unchanged; material — poplin is lighter, tighter-woven and lower in bending stiffness than the twill. Density 5 stitches/cm that the twill tolerated **jams the poplin's structure (displacement pucker)** and its low stiffness shows every tension effect. Remedy set applied together: density reduced to 4/cm, needle Nm 70, both tensions backed off

until the stitch just balances, presser pressure reduced. Test seams sewn, threads cut — fabric lies flat: confirmed. The lesson generalises: settings belong to the *fabric–machine combination*, never to the machine alone, which is why a settings record (Topic 10) is filled for every new style.

5. Prevention as a system

Factories that chase faults individually stay busy forever. The preventive system is: **needle policy** (scheduled changes, no re-use of dropped needles), **setting sheets** per style (machine, needle, thread, density, tensions, folder), **first-piece approval** at every operation start, **in-line inspection** catching faults at the operation that makes them, and a **fault log** analysed by Pareto — the vital few causes (typically needle damage, tension drift and ply shift) get engineering attention, not operator blame.

Glossary

TERM	MEANING
Flagging	Fabric lifting with the needle on its rise, disturbing loop formation on light materials.
Displacement pucker	Puckering caused by stitching crowding the fabric's own yarns; remains even if threads are removed.
Loop lift	The needle's rise after bottom dead centre (about 1.8–2.2 mm) at which the loop is presented to the hook.
Pareto analysis	Ranking fault causes by frequency so the few dominant causes are attacked first.

Key points and summary

- Diagnose in the cheap order: needle, re-threading, settings, material, handling.
- Cut the threads of a test seam to separate tension pucker from feed pucker before adjusting anything.
- Settings belong to the fabric–machine combination; every new style earns a fresh settings record.
- Prevention — needle policy, setting sheets, first-piece approval, Pareto review — beats fault-chasing.

Practice

Answer in your own words; brief model answers follow.

1. A lockstitch seam shows its knot sitting on the top surface. Which tension do you adjust first, in which direction, and why is the bobbin treated as the reference?

2. A seam is puckered; when its threads are cut out the two plies lie perfectly flat. Which pucker type is this, and what are the first two adjustments?
3. List four causes of skipped stitches and the check that confirms or clears each.
4. Why does changing only the fabric require a new settings record even when machine, needle and thread all stay the same?

Answer key

1. The knot is pulled up because needle-thread tension is too low relative to the bobbin (or the bobbin is too tight). Set/verify bobbin tension first as the stable reference, then raise needle tension until the interlacing buries mid-ply — adjusting the needle side alone against an unknown bobbin setting chases a moving target.
2. Tension pucker: the threads themselves were contracting the seam. First reduce needle (and bobbin) tensions to the minimum that balances; then reduce stitch density if needed.
3. Bent/blunt needle — replace and retest; needle too fine for the thread — check eye fit/size up; hook timing/loop lift off — gauge the 1.8–2.2 mm lift; fabric flagging — observe the fabric at the needle and test with firmer pressure or a smaller plate hole.
4. Because pucker behaviour, penetration force, feed behaviour and tension balance all arise from the fabric's weight, weave and stiffness interacting with the machine. The poplin case shows identical hardware producing a fault on one fabric and good work on another.
